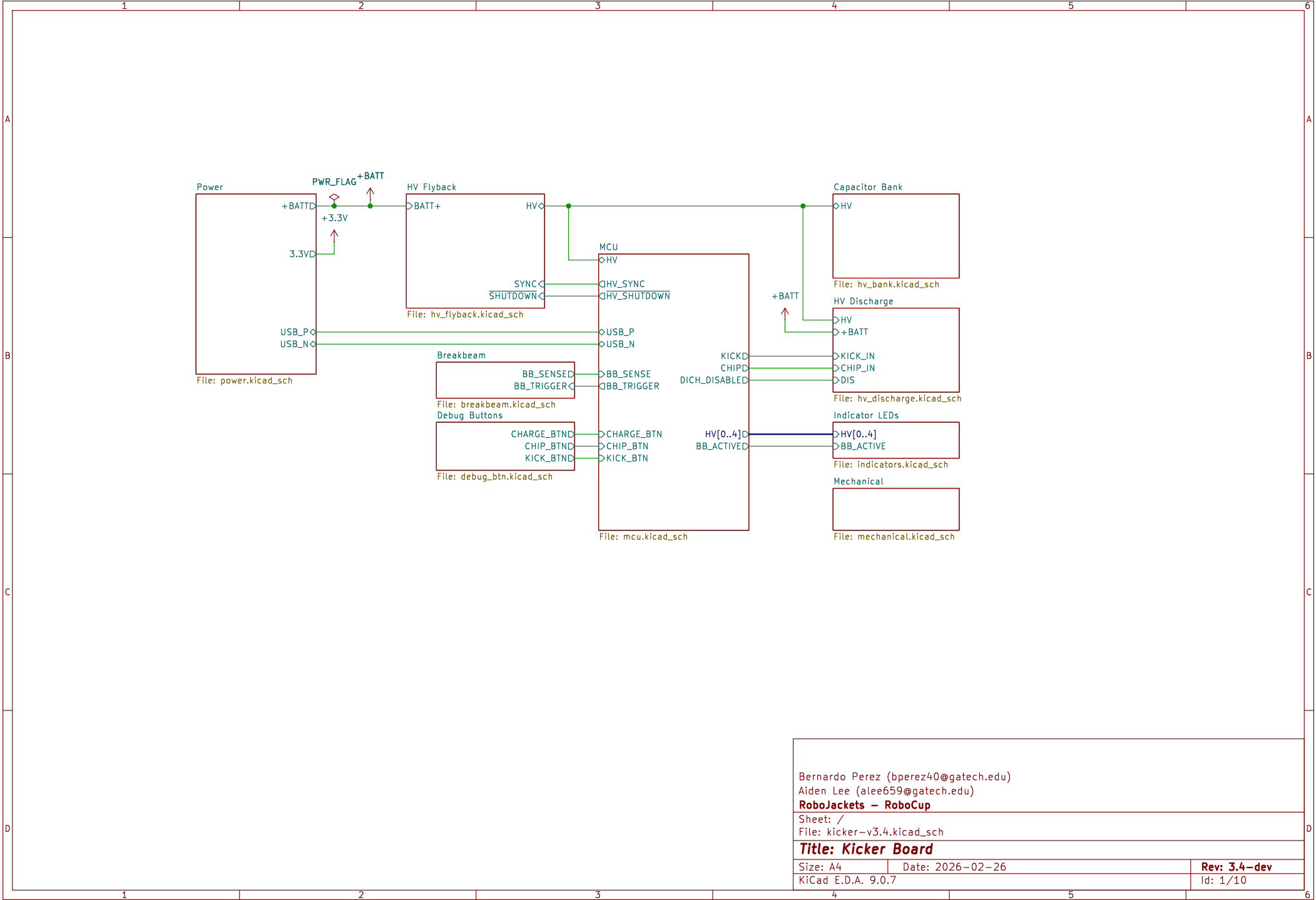




Bernardo Perez (bperez40@gatech.edu)

Aiden Lee (alee659@gatech.edu)

RoboJackets – RoboCup

Sheet: /

File: kicker-v3.4.kicad_sch

Title: Kicker Board

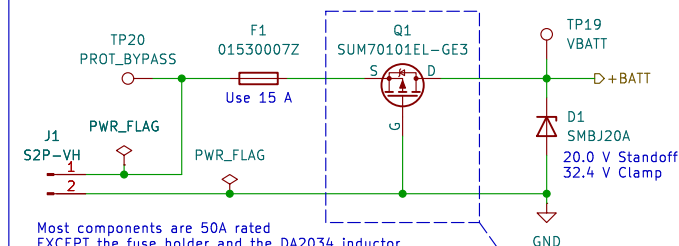
Size: A4 Date: 2026-02-26

KiCad E.D.A. 9.0.7

Rev: 3.4-dev

Id: 1/10

Power Input with Reverse Polarity Protection

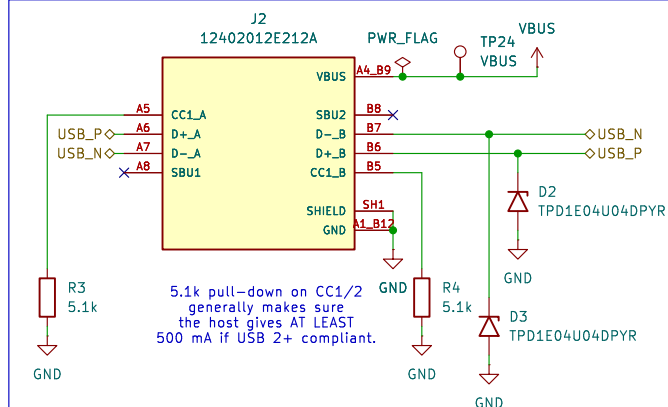


Most components are 50A rated EXCEPT the fuse holder and the DA2034 inductor. The inductor is rated for continuous operation of 10A, however the current design of the board does not operate at continuous current.

NOTE: Spec fuse based on the reverse current protection FET and diode on this sheet as well as the max switching current of Q1, T1, and D1 on the LT3751 sheet. The LT3751 current sense resistors should be set to limit current below the limit of the fuse.

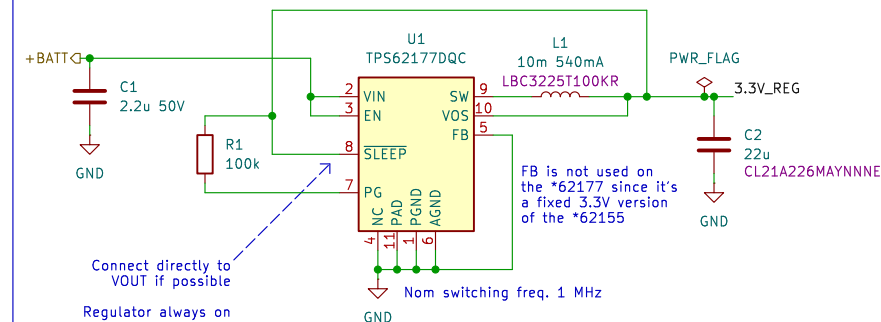
This PFET configuration is typical of a reverse polarity protection circuit. It isn't broken. There are a few blogs/videos on this just search "PFET reverse polarity protection".

USB-C Connector



5.1k pull-down on CC1/2 generally makes sure the host gives AT LEAST 500 mA if USB 2+ compliant.

3.3V 500 mA Switching Regulator

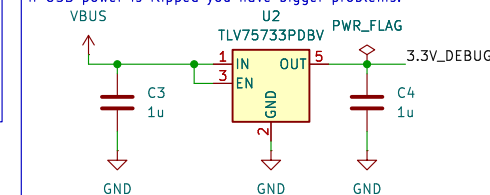


We use a switching regulator for +BATT to 3.3V because the LDOs are much less efficient than switching regulators for this voltage difference. ($\approx 16.5\%$ vs $>80\%$ for 20V +BATT)

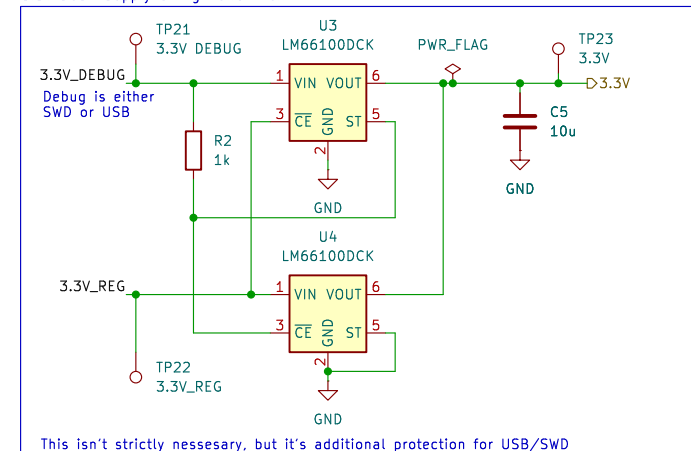
However, LDOs are acceptable for 5V to 3.3V step-down because this rail is only used for debugging and LDOs are far more efficient at this voltage drop ($\approx 66\%$).

3.3V 1A Low-Dropout Regulator (USB only)

Reverse polarity prot. not needed. If USB power is flipped you have bigger problems.



3.3V Dual-Supply ORing Power Mux



This isn't strictly necessary, but it's additional protection for USB/SWD

Expected current draw, 3.3V

Maximum Current Draw:
40 mA - RP2040 MCU
160 mA - 16 x LED @ 10 mA ea.
100 mA - Budget for Breakbeam

400 mA < 500 mA (Reg. Limit)

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Aiden Lee (alee659@gatech.edu)

RoboJackets - RoboCup

Sheet: /Power/

File: power.kicad_sch

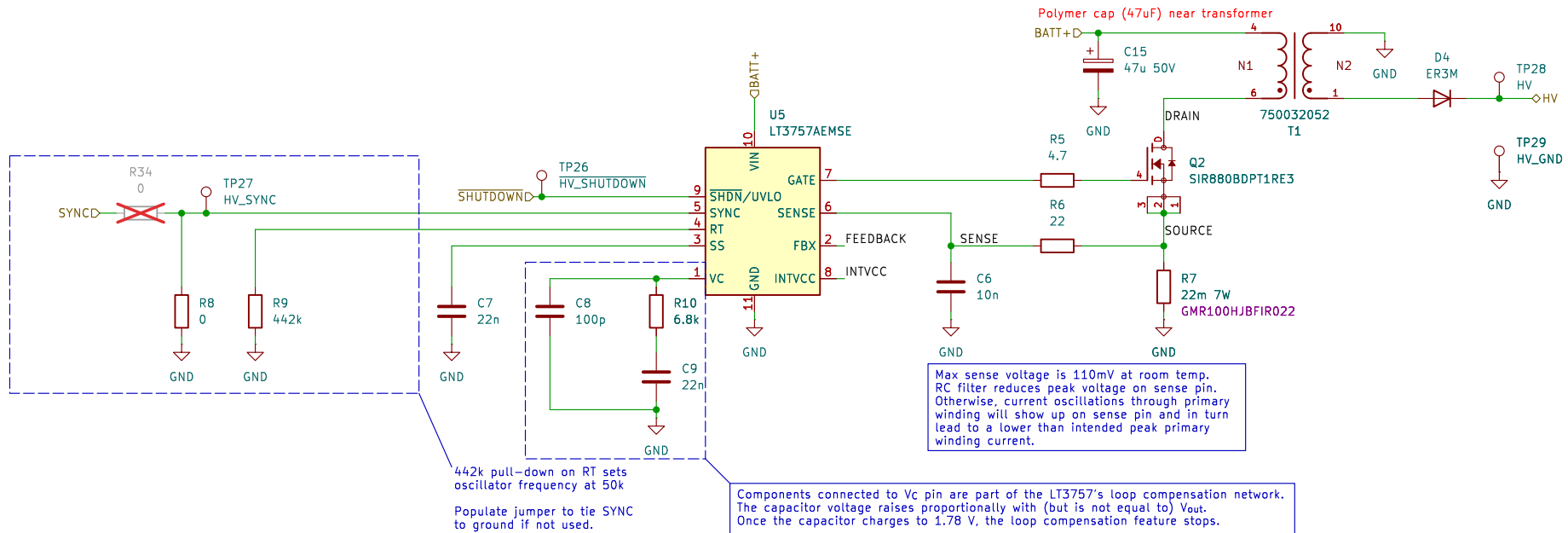
Title: Power Supply

Size: A4 Date: 2026-02-26

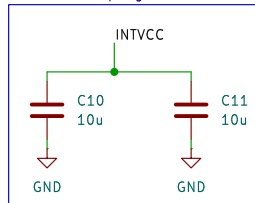
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Rev: 3.4-dev

Id: 2/10



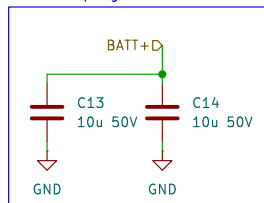
INTVCC Decoupling



INTVcc is supplied from Vin but still requires its own bypass capacitors.

Place caps close to INTVcc

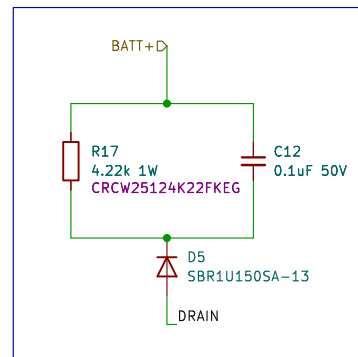
VIN Decoupling



Use combination of MLCC and polymer capacitors. Polymer capacitor serves as bulk decoupling. MLCC is high frequency decoupling.

Place MLCC (10uF) caps close to Vin pin.

RCD Snubber

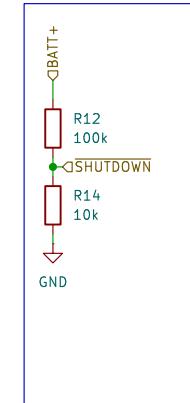


Res should be dissipating abt 1/4 W

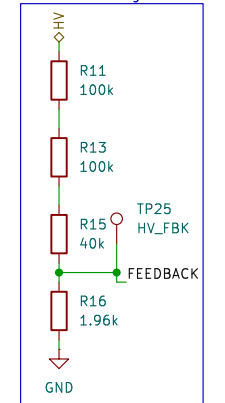
RCD snubber clamps MOSFET drain voltage. When diode become forward biased, capacitor charges, and resistor dissipates energy. RCD as opposed to RC allows lower Vds, thus we can select MOSFET with lower Rds(on)

From sim...
 $V_{sn} = 17.75 \text{ V}$
 $dV_{sn} = 890 \text{ mV (5\% of } V_{sn})$
 $V_{in(max)} = 22 \text{ V}$
 $V_{d(rvsmx)} = V_{sn} + V_{in(max)} = 40 \text{ V}$

UVLO Set



Feedback Voltage Divider



Target voltage dropped from 250V to 200V because Cap voltage dropped from 350V to 250V

$$V_{out} = 1.6 \cdot (R2/R1) = 1.6 \cdot 240k\Omega / 1.96k\Omega \sim 200V$$

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Aiden Lee (alee659@gatech.edu)

RoboJackets - RoboCup

Sheet: /HV Flyback/

File: hv_flyback.kicad_sch

Title: High Voltage Flyback Transformer

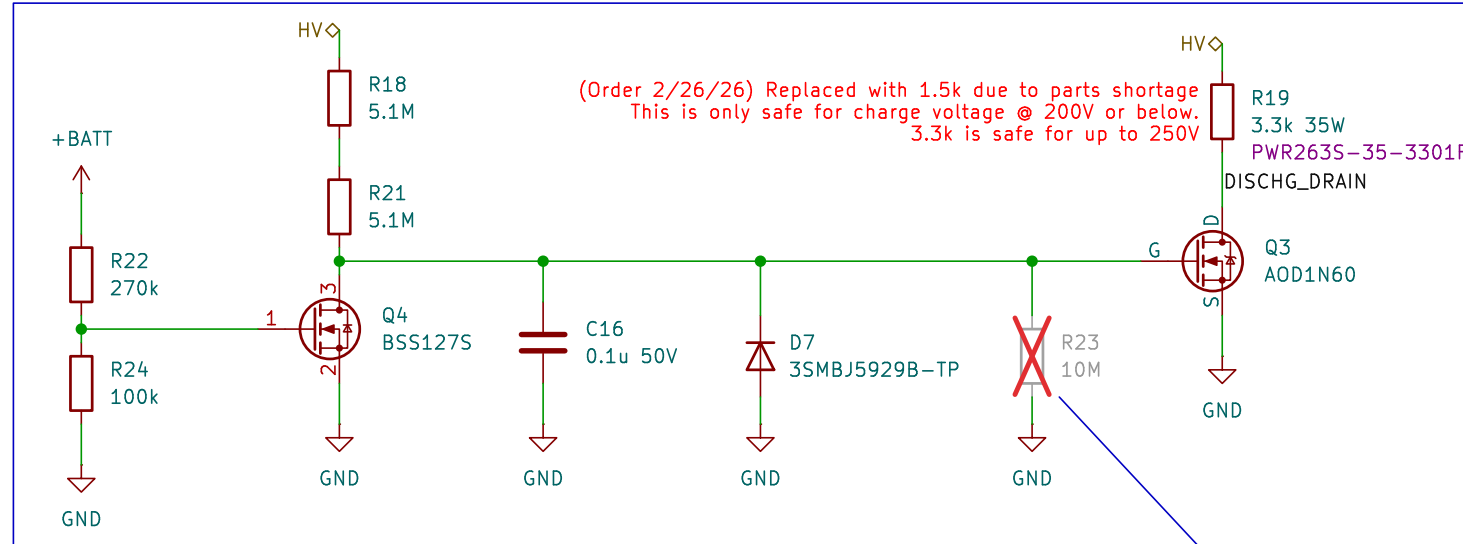
Size: A4 Date: 2026-02-26

KiCad E.D.A. 9.0.7

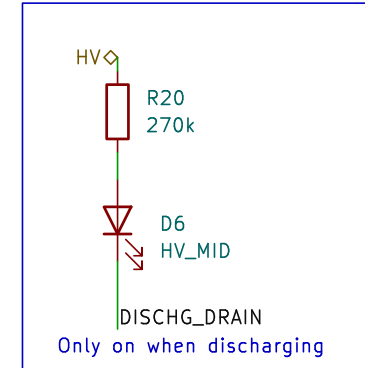
Rev: 3.4-dev

Id: 3/10

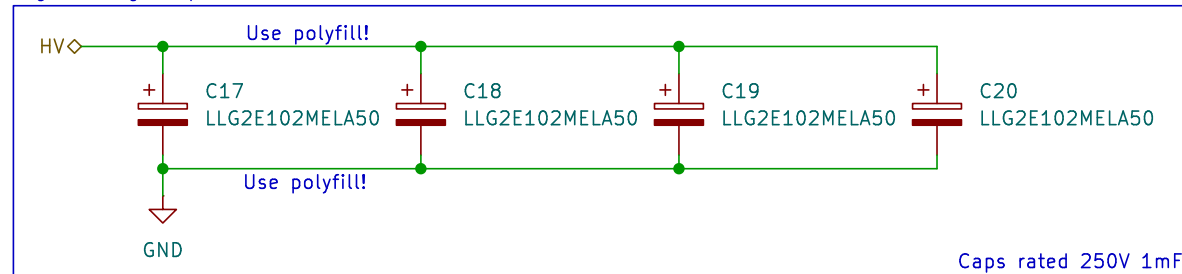
High-Voltage Auto Safety Discharge



Safety Discharge Indicator



High-Voltage Capacitor Bank



This is an optional pull-down resistor to create a voltage divider to prevent voltage spikes from damaging Q6 and should be left unpopulated unless needed.

Voltage Divider:

$V_{in_typ} = 250V$ (Typical max voltage)

$V_{out_typ} = 250(1.0E7 / (2 * 5.1E6 * 1.0E7)) = 123.8V$

$V_{in_spike} = 600V$

$V_{out_spike} = 600(1.0E7 / (2 * 5.1E6 + 1.0E7)) = 297V$

V_{th} of the AOD1N60 is 4.1V and serves as the limit for the lowest attainable voltage through the AOD alone (other factors notwithstanding). Adding a voltage divider changes the V_{min} seen on the HV line.

$V_{min} = V_{th}((2 * 5.1E6 + 1.0E7) / 1.0E7) = 8.28V$

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RoboJackets – RoboCup

Sheet: /Capacitor Bank/

File: hv_bank.kicad_sch

Title: High Voltage Cap Bank w/ Safety Auto Discharge

Size: A5

Date: 2026-02-26

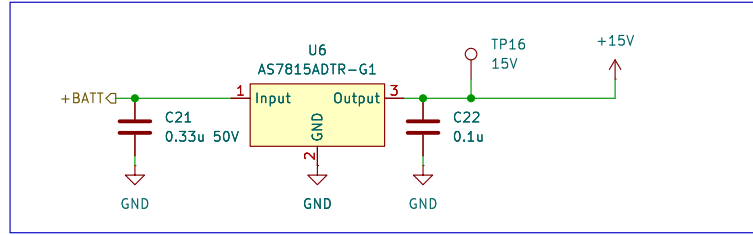
Rev: 3.4-dev

KiCad E.D.A. 9.0.7

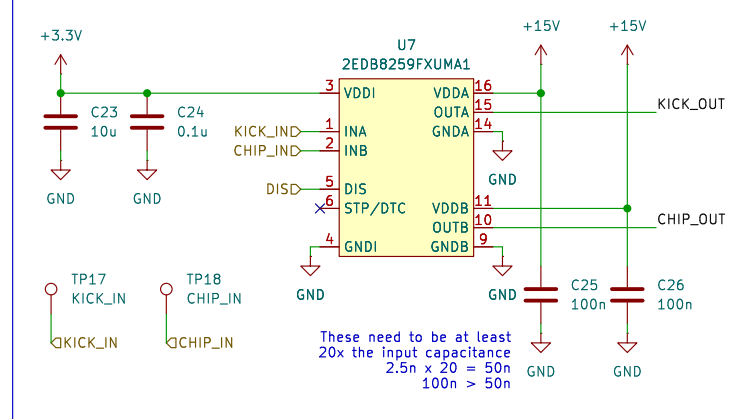
Id: 4/10

This is a copy of Will Stuckey's design with new components.
The previous design works in real-life testing

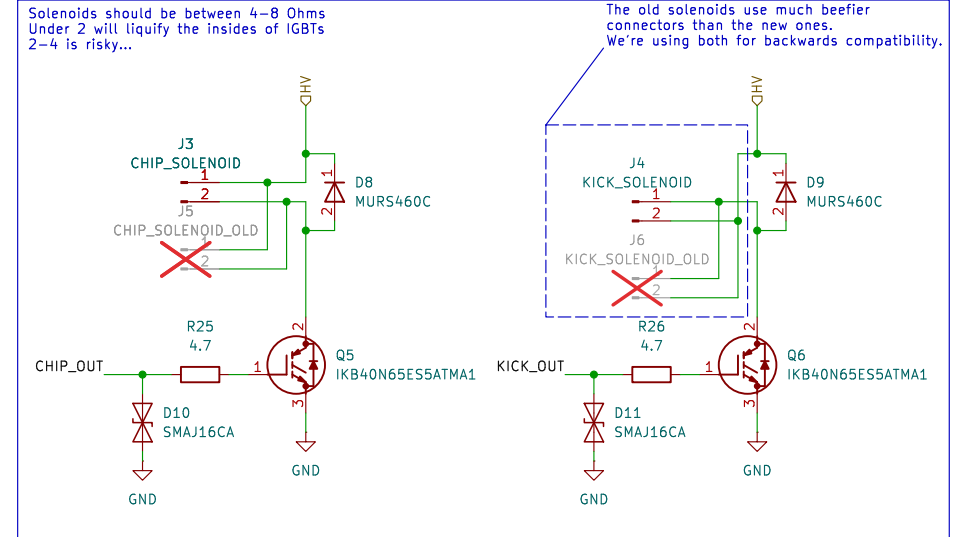
15V LDO for IGBT Gate Drive



IGBT Gate Drivers



Output IGBT



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Aiden Lee (alee659@gatech.edu)

RoboJackets – RoboCup

Sheet: /HV Discharge/

File: hv_discharge.kicad_sch

Title: High Voltage Discharge

Size: A4

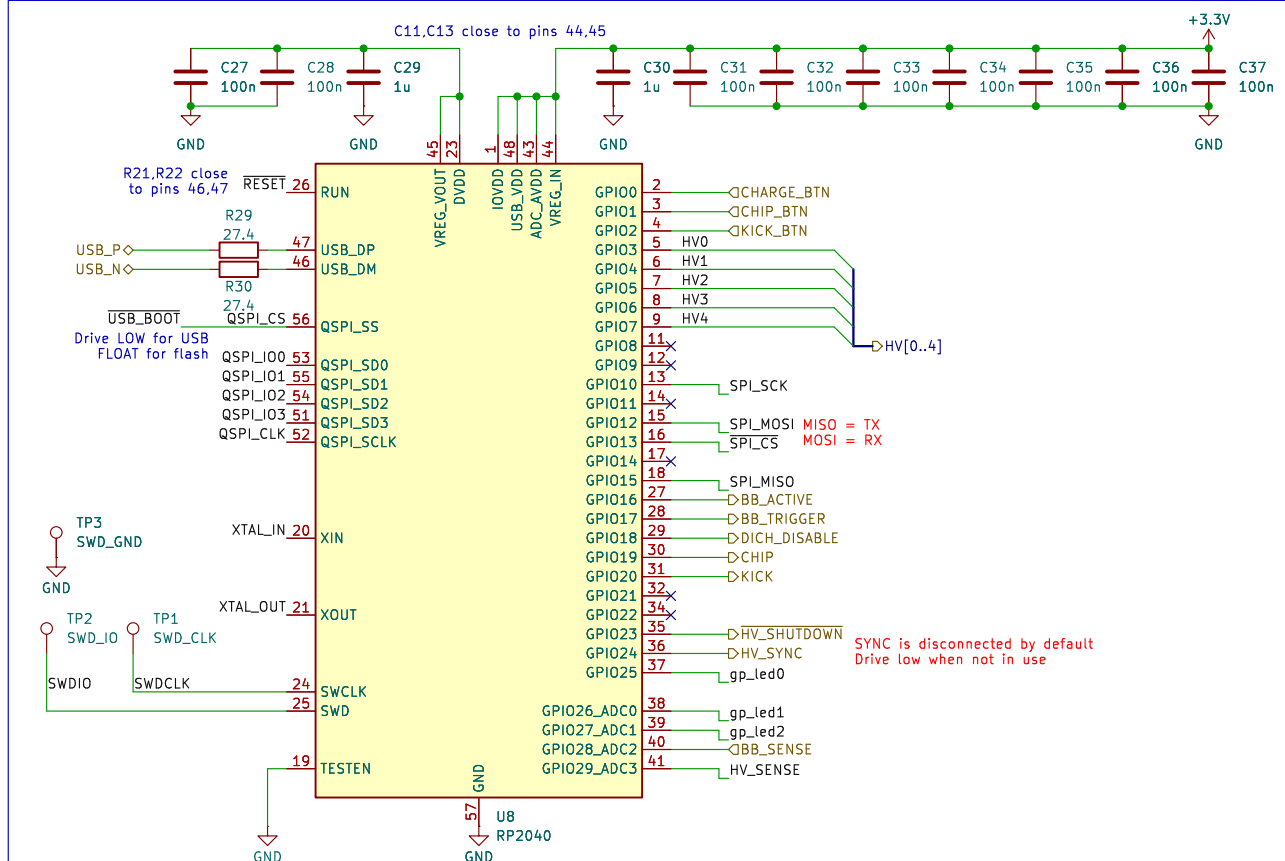
Date: 2026-02-26

Rev: 3.4-dev

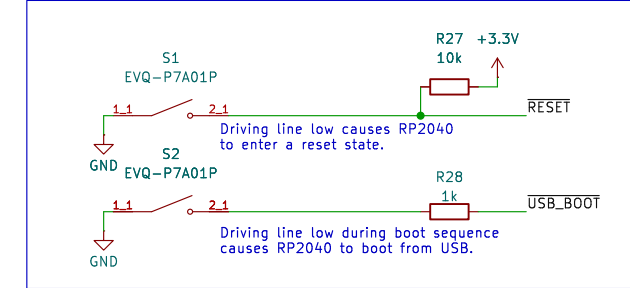
KiCad E.D.A. 9.0.7

Id: 5/10

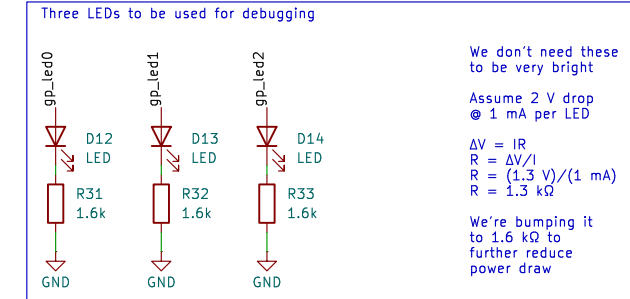
RP2040 MCU + Support Circuitry



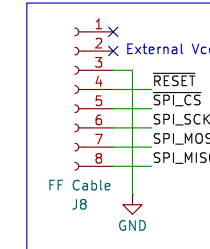
Debug Buttons



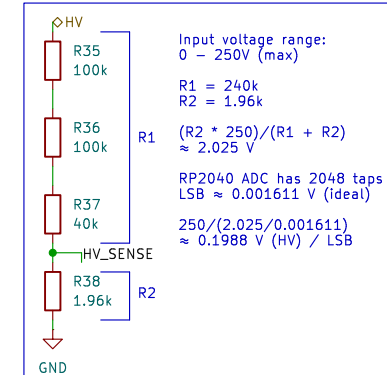
General-Purpose Indicator LEDs



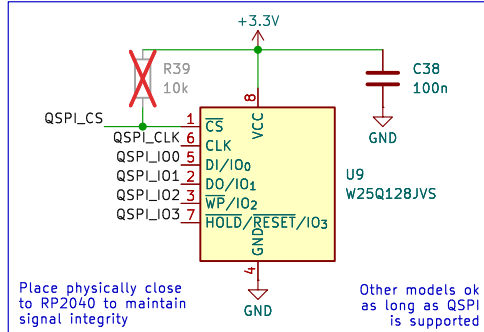
Data Connector



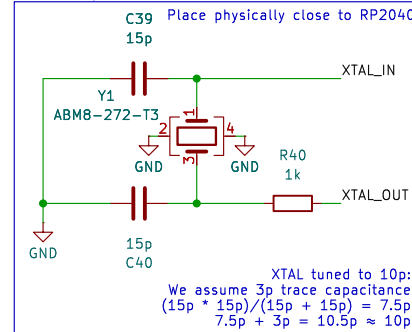
High Voltage Sense Line



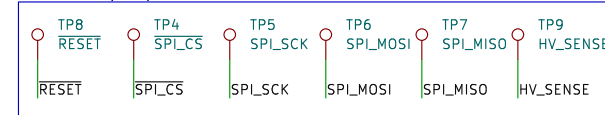
16 MB External SPI Flash



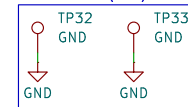
12 MHz Crystal



Test Points (Data)



Test Points (Gnd)



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RoboJackets - RoboCup

Sheet: /MCU/

File: mcu.kicad_sch

Title: RP2040 MCU + Support Circuitry

Size: A4 Date: 2026-02-26

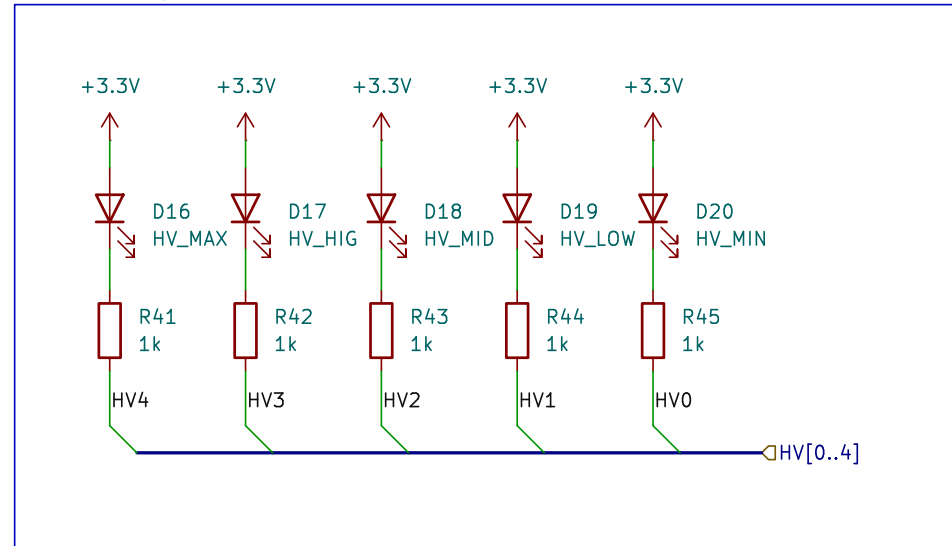
KiCad E.D.A. 9.0.7

Rev: 3.4-dev

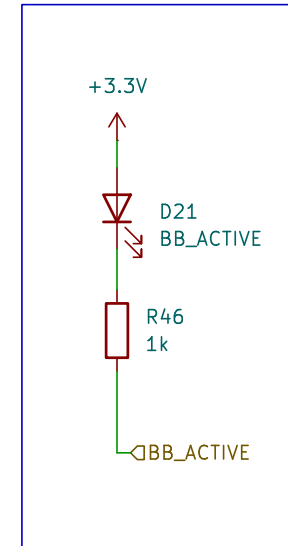
Id: 6/10

Design referenced from:
<https://datasheets.raspberrypi.com/rp2040/hardware-design-with-rp2040.pdf>

HV Bank Charge Status



Breakbeam status



Aiden Lee (alee659@gatech.edu)

RoboJackets – RoboCup

Sheet: /Indicator LEDs/

File: indicators.kicad_sch

Title: Indicator LEDs

Size: A5

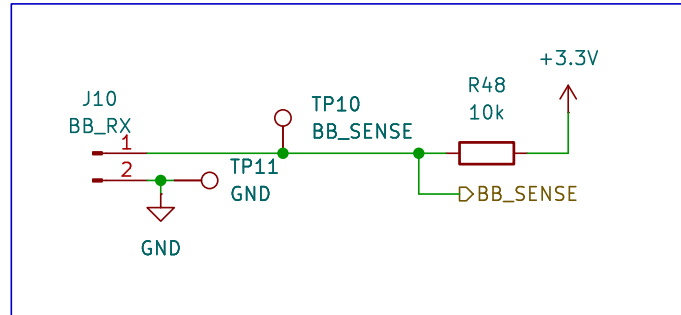
Date: 2026-02-26

Rev: 3.4-dev

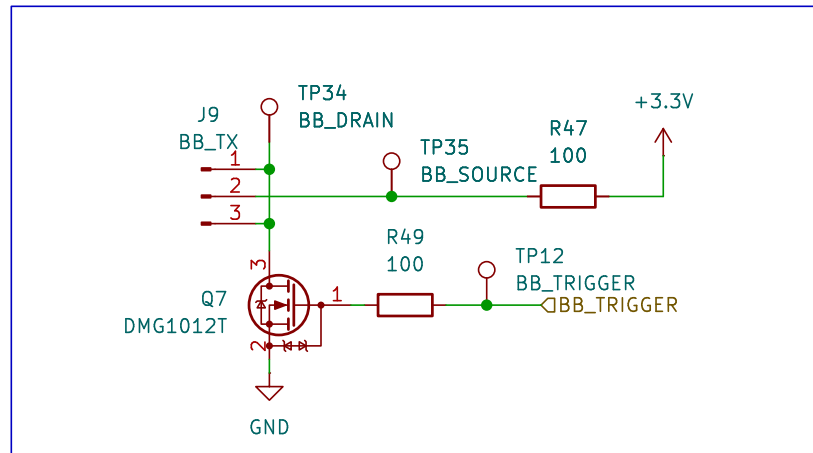
KiCad E.D.A. 9.0.7

Id: 7/10

Breakbeam Recieve



Breakbeam Transmit



Miles Schuler (mschuler9@gatech.edu)

Aiden Lee (alee659@gatech.edu)

RoboJackets – RoboCup

Sheet: /Breakbeam/

File: breakbeam.kicad_sch

Title: Laser Breakbeam Interface

Size: A5

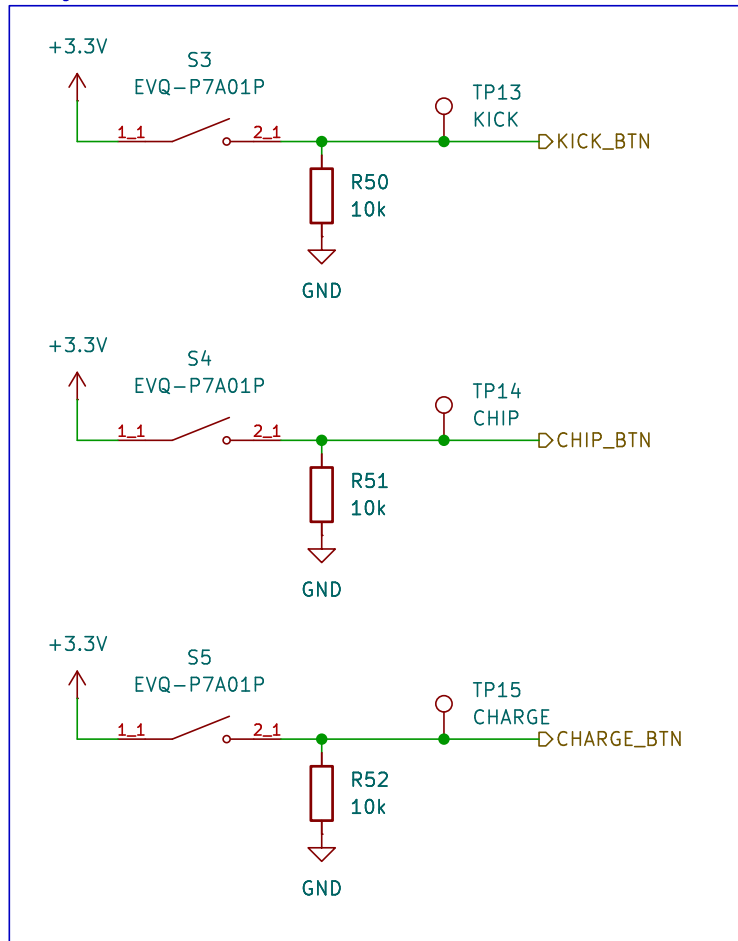
Date: 2026-02-26

Rev: 3.4-dev

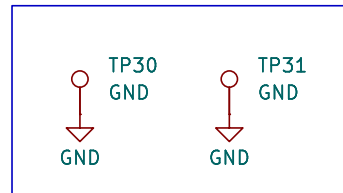
KiCad E.D.A. 9.0.7

Id: 8/10

Debug Switches



Extra Grounds



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RoboJackets – RoboCup

Sheet: /Debug Buttons/
File: debug_btn.kicad_sch

Title: Debug Switches

Size: A5

Date: 2026-02-26

Rev: 3.4-dev

KiCad E.D.A. 9.0.7

Id: 9/10



MountingHole_Pad



H2 MountingHole_Pad



MountingHole_Pad



H4 MountingHole_Pad



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RoboJackets – RoboCup

Sheet: /Mechanical/

File: mechanical.kicad_sch

Title: Mechanical Parts

Size: A5

Date: 2026-02-26

Rev: 3.4-dev

KiCad E.D.A. 9.0.7

Id: 10/10